The University of Jordan (UJ)
School of Engineering
Department of Computer Engineering
Advanced Networks Lab 0907529
Exp.4 Introduction to Wireless LANs

Objectives

1. Review types of wireless networks.
2. Discuss 802.11 standards.
3. Demonstrate WLAN components.
4. Illustrate wireless topology modes.
5. Illustrate the process of wireless client and AP Association.
6. Configure WLAN

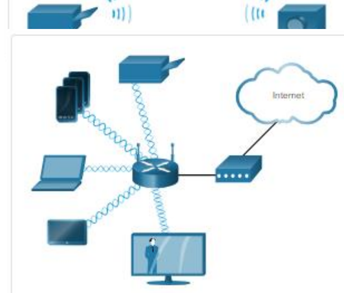
Introduction

A Wireless LAN (WLAN) is a type of wireless network that is commonly used in homes, offices, and campus environments. Networks must support people who are on the move. People connect using computers, laptops, tablets, and smart phones. There are many different network infrastructures that provide network access, such as wired LANs, service provider networks, and cell phone networks. But it's the WLAN that makes mobility possible within the home and business environments.

Types of Wireless Networks

Wireless networks are based on the Institute of Electrical and Electronics Engineers (IEEE) standards and can be classified broadly into four main types: WPAN, WLAN, WMAN, and WWAN.

Wireless Personal-Area Networks (WPAN) - Uses low powered transmitters for a short-range network, usually 20 to 30 ft. (6 to 9 meters). Bluetooth and ZigBee based devices are commonly used in WPANs. WPANs are based on the 802.15 standard and a 2.4-GHz radio frequency.



Wireless LANs (WLAN) - Uses transmitters to cover a medium-sized network, usually up to 300 feet. WLANs are suitable for use in a home, office, and even a campus environment. WLANs are based on the 802.11 standard and a 2.4-GHz or 5-GHz radio frequency.

Wireless MANs (WMAN) - Uses transmitters to provide wireless service over a larger geographic area. WMANs are suitable for providing wireless access to a metropolitan city or specific district. WMANs use specific licensed frequencies.

Wireless Wide-Area Networks (WWANs) - Uses transmitters to provide coverage over an extensive geographic area. WWANs are suitable for national and global communications. WWANs also use specific licensed frequencies.

<u>802.11 Standards</u>

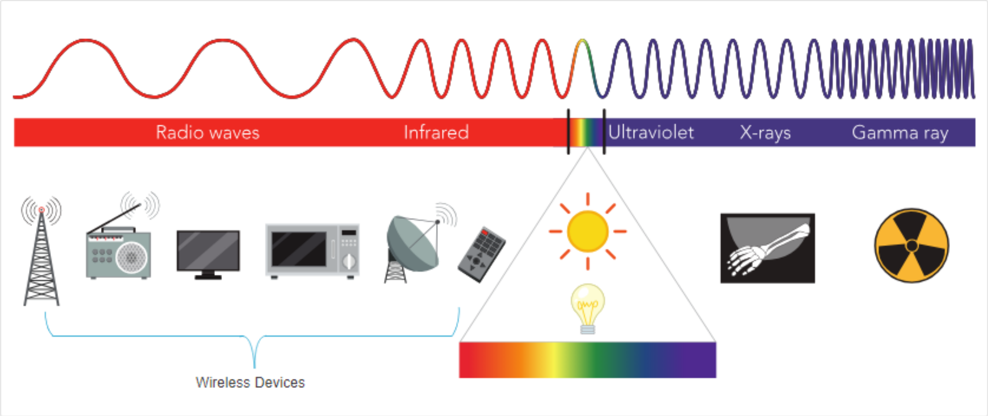
Various implementations of the IEEE 802.11 standard have been developed over the years. The table highlights these standards.

IEEE WLAN Standard	Radio Frequency	Description
802.11	2.4 GHz	• speeds of up to 2 Mbps
802.11a	5 GHz	• speeds of up to 54 Mbps • small coverage area • less effective at penetrating building structures • not interoperable with the 802.11b and 802.11g
802.11b	2.4 GHz	• speeds of up to 11 Mbps • longer range than 802.11a • better able to penetrate building structures
802.11g	2.4 GHz	• speeds of up to 54 Mbps • backward compatible with 802.11b with reduced bandwidth capacity
802.11n	2.4 GHz 5 GHz	• data rates range from 150 Mbps to 600 Mbps with a distance range of up to 70 m (230 feet) • APs and wireless clients require multiple antennas using MIMO technology • backward compatible with 802.11a/b/g devices with limiting data rates
802.11ac	5 GHz	• provides data rates ranging from 450 Mbps to 1.3 Gbps (1300 Mbps) using MIMO technology • Up to eight antennas can be supported • backwards compatible with 802.11a/n devices with limiting data rates
802.11ax	2.4 GHz 5 GHz	• latest standard released in 2019 • also known as Wi-Fi 6 or High-Efficiency Wireless (HEW)

IEEE WLAN Standard	Radio Frequency	Description
		• provides improved power efficiency, higher data rates, increased capacity, and handles many connected devices • currently operates using 2.4 GHz and 5 GHz but will use 1 GHz and 7 GHz when those frequencies become available • Search the internet for Wi-Fi Generation 6 for more information

All wireless devices operate in the radio waves range of the electromagnetic spectrum. WLAN networks operate in the 2.4 GHz frequency band and the 5 GHz band. Wireless LAN devices have transmitters and receivers tuned to specific frequencies of the radio waves range, as shown in the figure. Specifically, the following frequency bands are allocated to 802.11 wireless LANs:

- 2.4 GHz (UHF) - 802.11b/g/n/ax
- 5 GHz (SHF) - 802.11a/n/ac/ax



WLAN Components

The physical WLAN architecture is fairly simple. Basic components of WLAN are typically: network interface cards (NICs) or client adaptors, wireless home routers, and wireless access points. You can use other components, such as wireless bridges and repeaters, to extend the reach of your network.

Wireless NICs - Wireless deployments require a minimum of two devices that have a radio transmitter and a radio receiver tuned to the same radio frequencies:

- End devices with wireless NICs
- A network device, such as a wireless router or wireless AP

To communicate wirelessly, laptops, tablets, smart phones, and even the latest automobiles include integrated wireless NICs that incorporate a radio transmitter/receiver. However, if a device does not have an integrated wireless NIC, then a USB wireless adapter can be used, as shown in the figure.

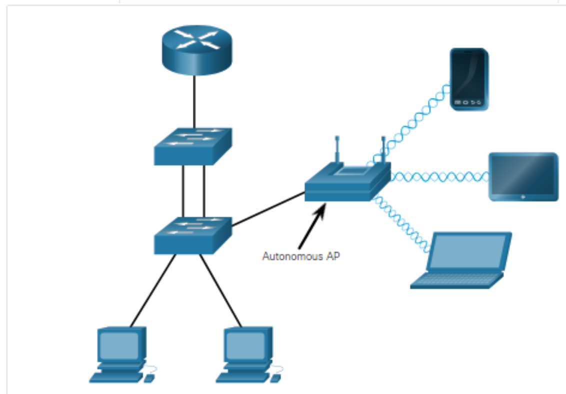
Wireless Home Router - The type of infrastructure device that an end device associates and authenticates with varies based on the size and requirement of the WLAN. For example, a home user typically interconnects wireless devices using a small, wireless router, as shown in the figure. The wireless router serves as an:

- Access point - This provides 802.11a/b/g/n/ac wireless access.
- Switch - This provides a four-port, full-duplex, 10/100/1000 Ethernet switch to interconnect wired devices.
- Router - This provides a default gateway for connecting to other network infrastructures, such as the internet.

Wireless Access Points - While range extenders are easy to set up and configure, the best solution would be to install another wireless access point to provide dedicated wireless access to the user devices. Wireless clients use their wireless NIC to discover nearby APs advertising their SSID. Clients then attempt to associate and authenticate with an AP. After being authenticated, wireless users have access to network resources. The Cisco Meraki Go APs are shown in the figure.

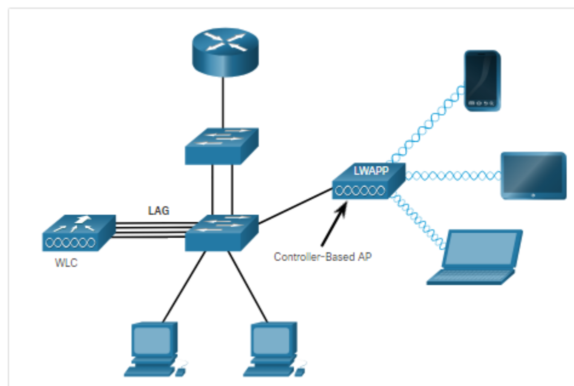
APs can be categorized as either autonomous APs or controller-based APs.

1. Autonomous APs



These are standalone devices configured using a command line interface or a GUI, as shown in the figure. Autonomous APs are useful in situations where only a couple of APs are required in the organization. A home router is an example of an autonomous AP because the entire AP configuration resides on the device. If the wireless demands increase, more APs would be required. Each AP would operate independent of other APs and each AP would require manual configuration and management. This would become overwhelming if many APs were needed.

2. Controller-based APs



These devices require no initial configuration and are often called lightweight APs (LAPs). LAPs use the Lightweight Access Point Protocol (LWAPP) to communicate with a WLAN controller (WLC), as shown in the next figure. Controller-based APs are useful in situations where many APs are required in the network. As more APs are added, each AP is automatically configured and managed by the WLC.

Most business class APs require external antennas to make them fully functioning units.

<u>Omnidirectional antennas</u> such as the one shown in the figure provide 360-degree coverage and are ideal in houses, open office areas, conference rooms, and outside areas.

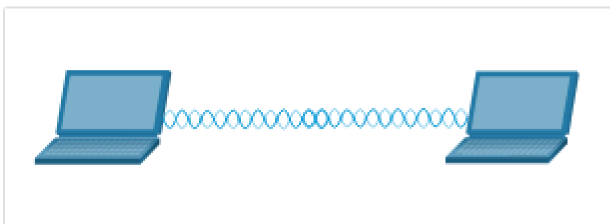
Directional antennas focus the radio signal in a given direction. This enhances the signal to and from the AP in the direction the antenna is pointing This provides a stronger signal strength in one direction and reduced signal strength in all other directions. Examples of directional Wi-Fi antennas include Yagi and parabolic dish antennas.

Multiple Input Multiple Output (MIMO) uses multiple antennas to increase available bandwidth for IEEE 802.11n/ac/ax wireless networks. Up to eight transmit and receive antennas can be used to increase throughput.

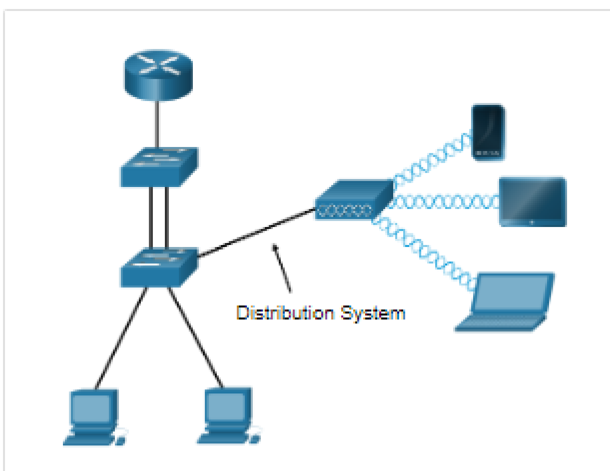
Wireless Topology Modes

Wireless LANs can accommodate various network topologies. The 802.11 standard identifies two main wireless topology modes: Ad hoc mode and Infrastructure mode.

Ad hoc mode - This is when two devices connect wirelessly in a peer-to-peer (P2P) manner without using APs or wireless routers. Examples include wireless clients connecting directly to each other using Bluetooth or Wi-Fi Direct. The IEEE 802.11 standard refers to an ad hoc network as an independent basic service set (IBSS).



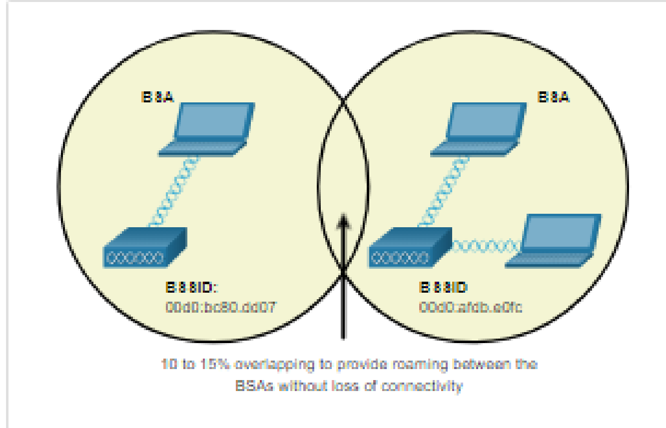
Infrastructure mode - This is when wireless clients interconnect via a wireless router or AP, such as in WLANs. APs connect to the network infrastructure using the wired distribution system, such as Ethernet.



Infrastructure mode defines two topology building blocks: A Basic Service Set (BSS) and an Extended Service Set (ESS).

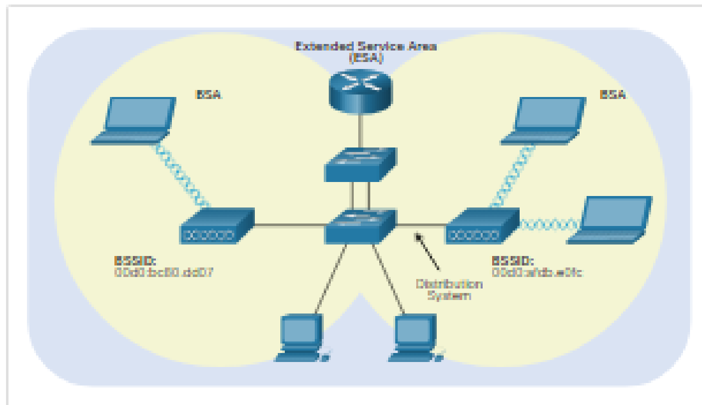
Basic Service Set - A BSS consists of a single AP interconnecting all associated wireless clients. Two BSSs are shown in the figure. The circles depict the coverage area for the BSS, which is called the Basic Service Area (BSA). If a wireless client moves out of its BSA, it can no longer directly communicate with other wireless clients within the BSA.

The Layer 2 MAC address of the AP is used to uniquely identify each BSS, which is called the Basic Service Set Identifier (BSSID). Therefore, the BSSID is the formal name of the BSS and is always associated with only one AP.



Extended Service Set - When a single BSS provides insufficient coverage, two or more BSSs can be joined through a common distribution system (DS) into an ESS. An ESS is the union of two or more BSSs interconnected by a wired DS. Each ESS is identified by a SSID and each BSS is identified by its BSSID.

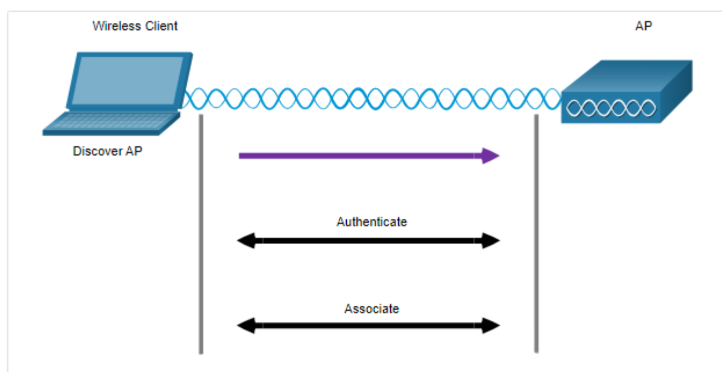
Wireless clients in one BSA can now communicate with wireless clients in another BSA within the same ESS. Roaming mobile wireless clients may move from one BSA to another (within the same ESS) and seamlessly connect. The rectangular area in the figure depicts the coverage area within which members of an ESS may communicate. This area is called the Extended Service Area (ESA).



Wireless Client and AP Association

For wireless devices to communicate over a network, they must first associate with an AP or wireless router. An important part of the 802.11 process is discovering a WLAN and subsequently connecting to it. Wireless devices complete the following three stage process, as shown in the figure:

- Discover a wireless AP
- Authenticate with AP
- Associate with AP



In order to have a successful association, a wireless client and an AP must agree on specific parameters. Parameters must then be configured on the AP and subsequently on the client to enable the negotiation of a successful association.

SSID -The SSID name appears in the list of available wireless networks on a client. In larger organizations that use multiple VLANs to segment traffic, each SSID is mapped to one VLAN. Depending on the network configuration, several APs on a network can share a common SSID.

Password - This is required from the wireless client to authenticate to the AP.

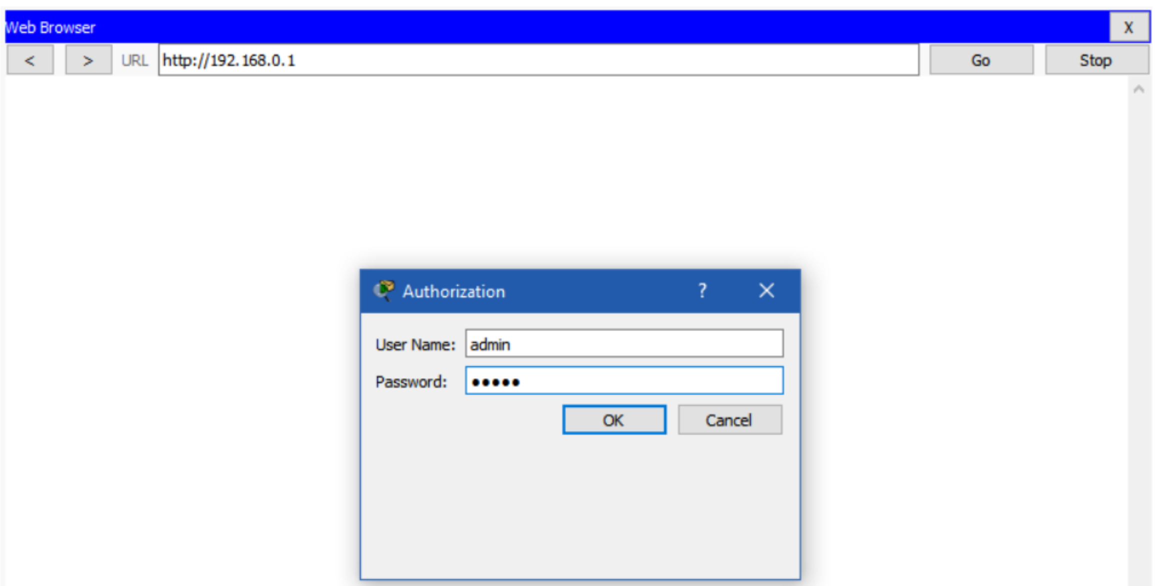
Network mode - This refers to the 802.11a/b/g/n/ac/ad WLAN standards. APs and wireless routers can operate in a Mixed mode meaning that they can simultaneously support clients connecting via multiple standards.

Security mode - This refers to the security parameter settings, such as WEP, WPA, or WPA2. Always enable the highest security level supported.

Channel settings - This refers to the frequency bands used to transmit wireless data. Wireless routers and APs can scan the radio frequency channels and automatically select an appropriate channel setting. The channel can also be set manually if there is interference with another AP or wireless device.

Basic Wireless Configuration

Most wireless routers are ready for service out of the box. They are preconfigured to be connected to the network and provide services. For example, the wireless router uses DHCP to automatically provide addressing information to connected devices. However, wireless router default IP addresses, usernames, and passwords can easily be found on the internet. Just enter the search phrase "default wireless router IP address" or "default wireless router passwords" to see a listing of many websites that provide this information. For example, username and password for the wireless router in the figure is "admin". Therefore, your first priority should be to change these defaults for security reasons.



To gain access to the wireless router's configuration GUI, open a web browser. In the address field, enter the default IP address for your wireless router. The default IP address can be found in the documentation that came with the wireless router or you can search the internet. The figure above shows the IPv4 address 192.168.0.1, which is a common default for many manufacturers. A security window prompts for authorization to access the router GUI. The word admin is commonly used as the default username and password. Again, check your wireless router's documentation or search the internet.

Basic network setup includes the following steps: 1

1. Log in to the router from a web browser.
2. Change the default administrative password.
3. Log in with the new administrative password.
4. Change the default DHCP IPv4 addresses.
5. Renew the IP address.
6. Log in to the router with the new IP address.

Laptop :- → wireless connection
→ wireless port
→ physical
→ port
أهمه راح ب
ال ethernet وحط port
wireless وبنشغل الجهاز

Basic wireless network configuration

Basic wireless configuration includes the following steps:

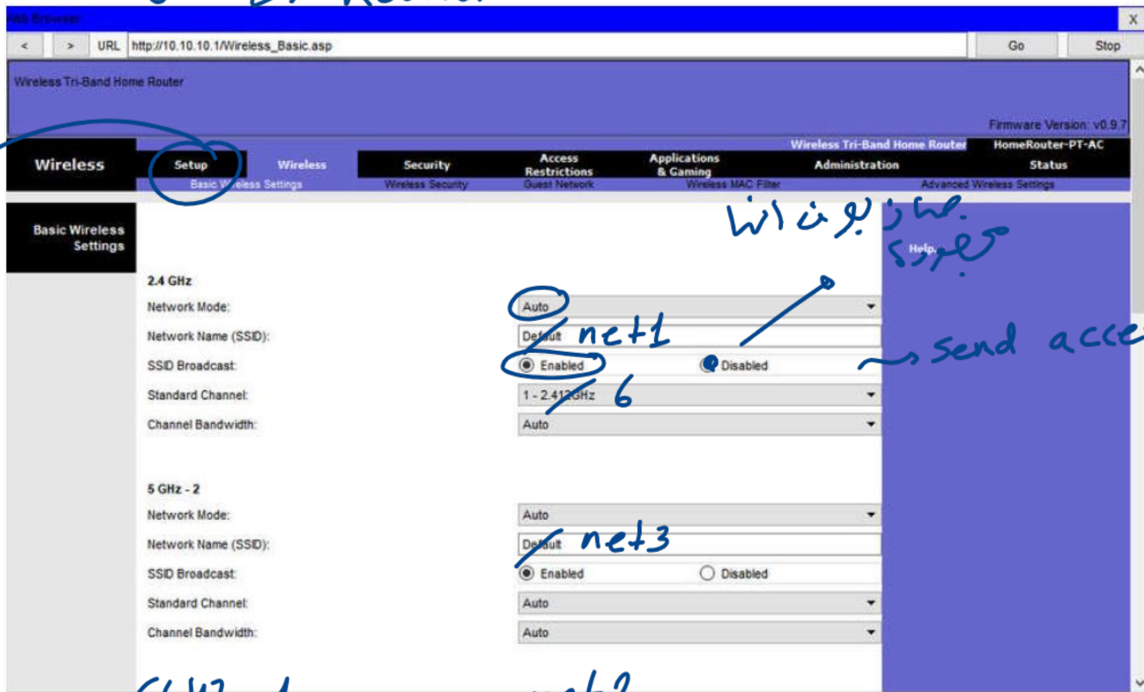
1. View the WLAN defaults.
2. Change the network mode.
3. Configure the SSID.
4. Configure the channel.
5. Configure the security mode.
6. Configure the passphrase.

1. View the WLAN defaults.

Out of the box, a wireless router provides wireless access to devices using a default wireless network name and password. The network name is called the Service Set Identified (SSID). Locate the basic wireless settings for your router to change these defaults, as shown in the example.

wireless Router

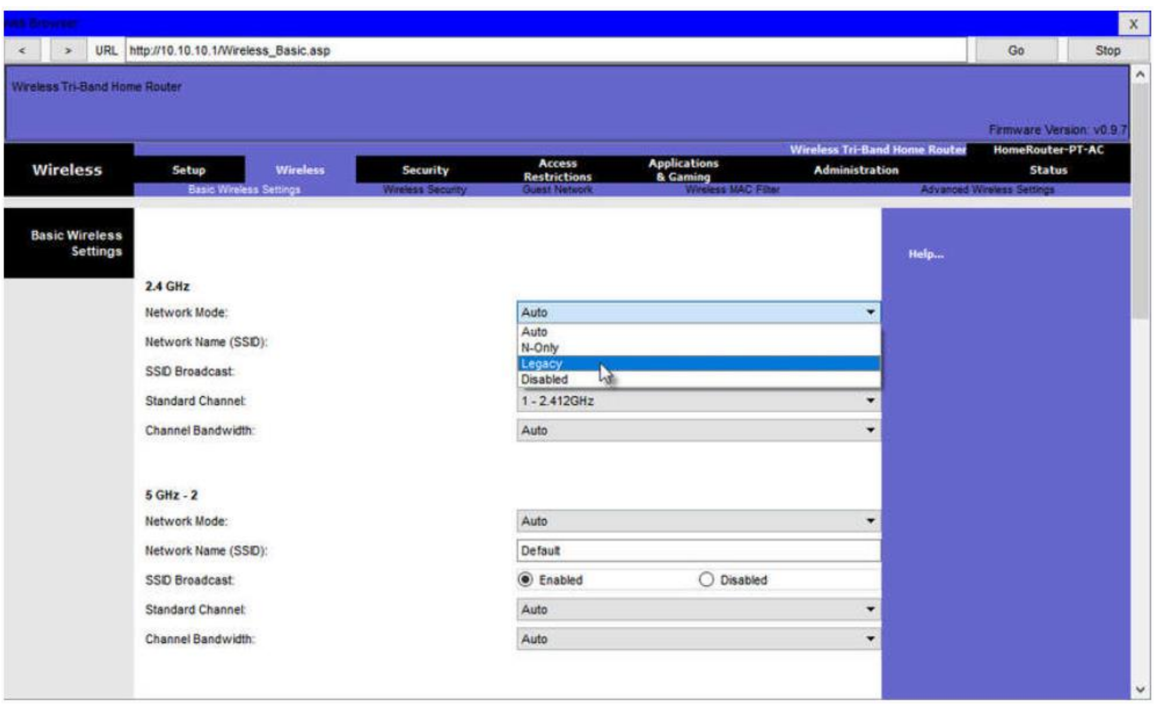
DHCP ال Ip addr



save settings

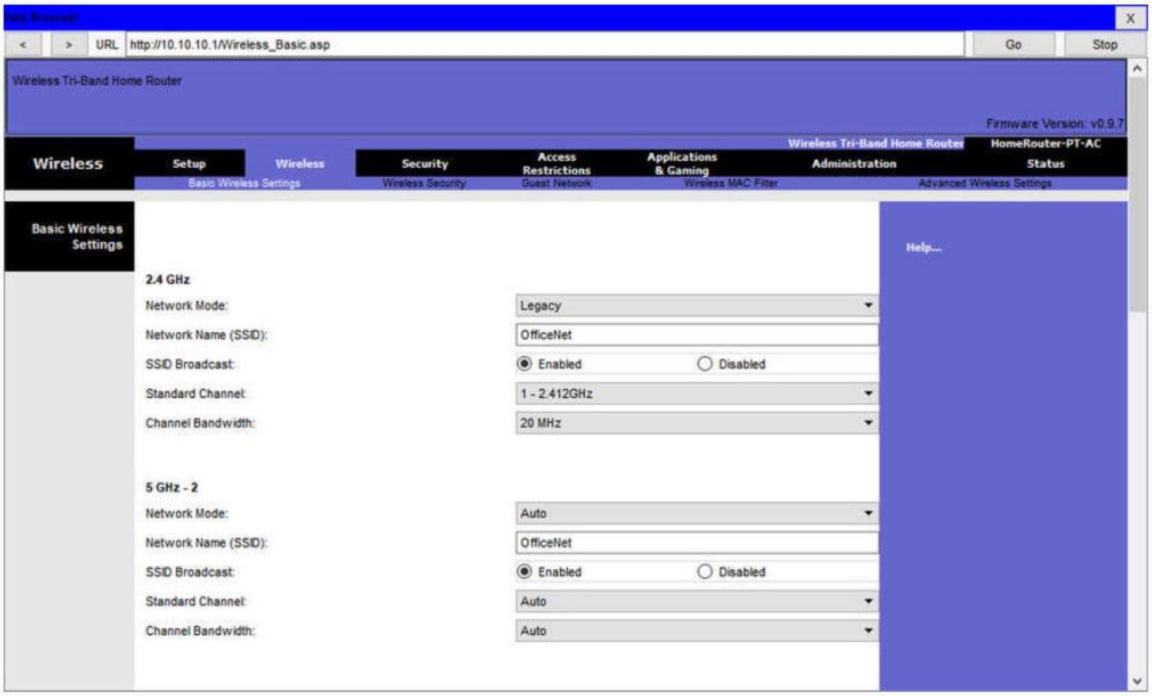
2. Change the network mode.

Some wireless routers allow you to select which 802.11 standard to implement. The example shows that "Legacy" has been selected. This means wireless devices connecting to the wireless router can have a variety of wireless NICs installed. Today's wireless routers configured for legacy or mixed mode most likely support 802.11a, 802.11n, and 802.11ac NICs.



3. Configure the SSID.

Assign an SSID to the WLANs. OfficeNet is used in the example for all three WLANs (the third WLAN is not shown). The wireless router announces its presence by sending broadcasts advertising its SSID. This allows wireless hosts to automatically discover the name of the wireless network. If the SSID broadcast is disabled, you must manually enter the SSID on each wireless device that connects to the WLAN.

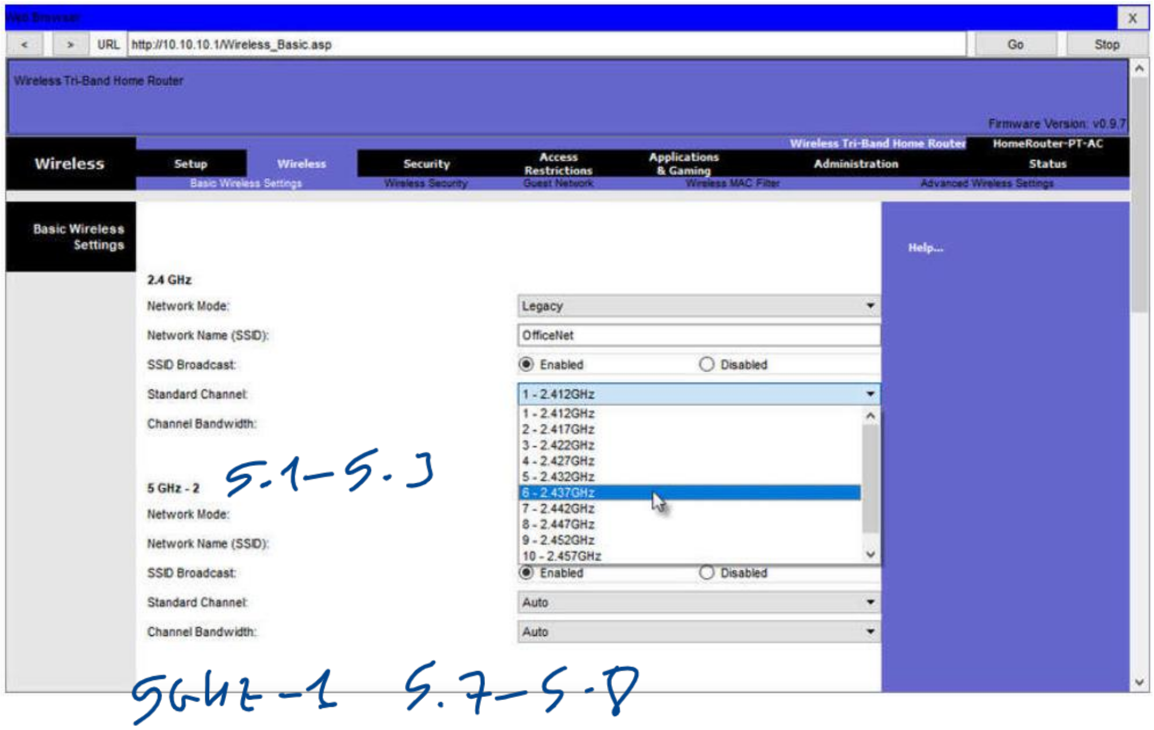


4. Configure the channel.

Devices configured with the same channel within the 2.4GHz band may overlap and cause distortion, slowing down the wireless performance and potentially break network connections. The solution to avoid interference is to configure non-overlapping channels on the wireless routers and access points that are near to each other. Specifically, channels 1, 6, and 11 are non-overlapping. In the example, the wireless router is configured to use channel 6.

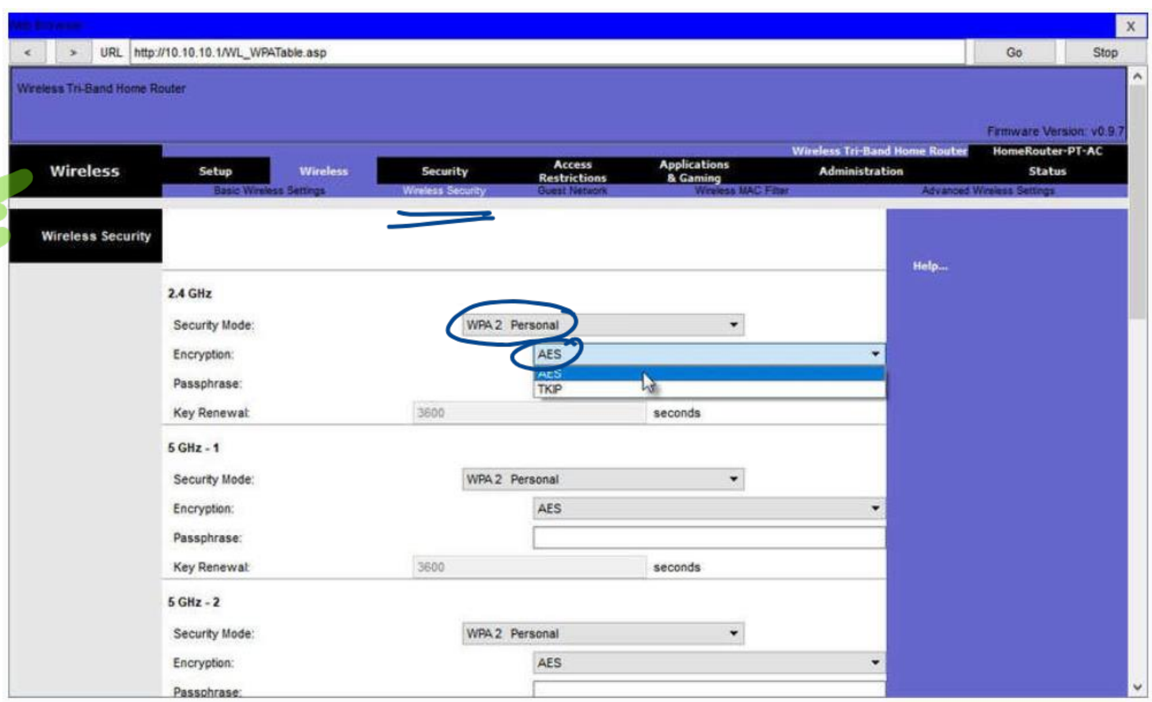
3

→ laptop
→ config
→ wireless
→ Desktop → pc wireless → connect
→ Refresh → pick one → connect
→ Connection ال رجع
Link Information شوف
Encryption policy.



5. Configure the security mode.

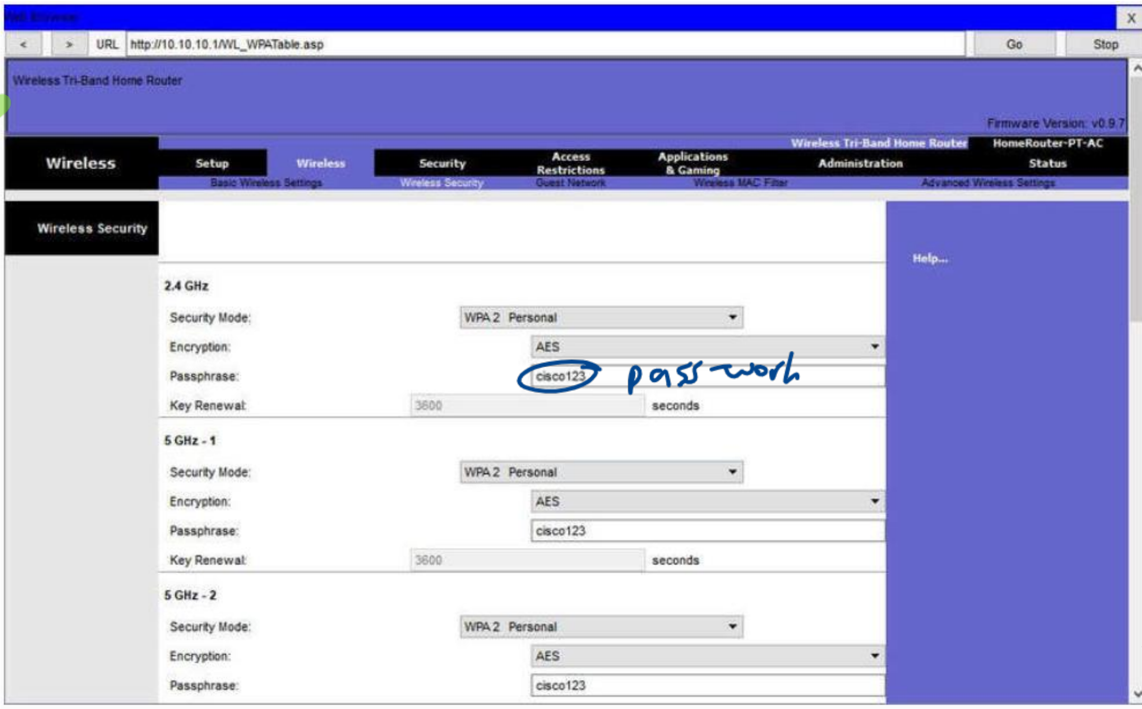
Out of the box, a wireless router may have no WLAN security configured. In the example, the personal version of Wi-Fi Protected Access version 2 (WPA2 Personal) is selected for all three WLANs. WPA2 with Advanced Encryption Standard (AES) encryption is currently the strongest security mode.



6. Configure the passphrase.

WPA2 personal uses a passphrase to authenticate wireless clients. WPA2 personal is easier to use in a small office or home environment because it does not require an authentication server. Larger organizations implement WPA2 enterprise and require wireless clients to authenticate with a username and password.

5



→ save settings!.

Configure a Basic WLAN on the WLC

6

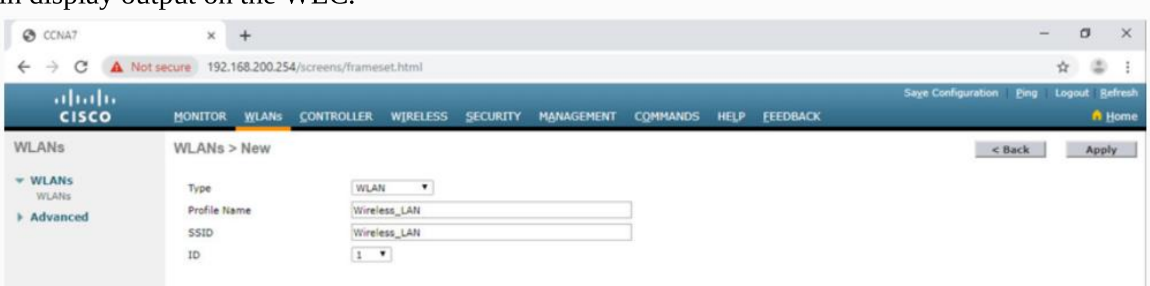
Basic WLAN configuration on the WLC includes the following steps:

1. Create the WLAN
2. Apply and Enable the WLAN
3. Select the Interface
4. Secure the WLAN
5. Verify the WLAN is Operational
6. Monitor the WLAN
7. View Wireless Client Information

laptop
→ connect → Refresh → net2
→ password & same encryption

1. Create the WLAN

In the figure, the administrator is creating a new WLAN that will use **Wireless_LAN** as the name and service set identifier (SSID). The ID is an arbitrary value that is used to identify the WLAN in display output on the WLC.



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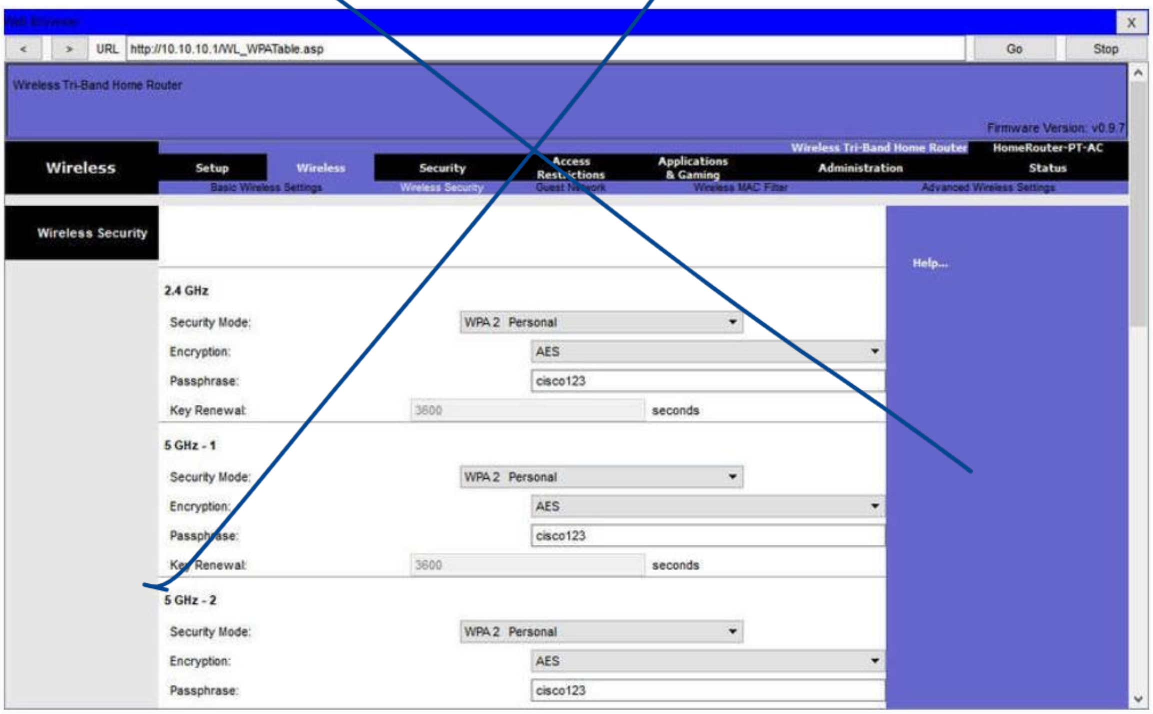


Configure a Basic WLAN on the WLC

*part2

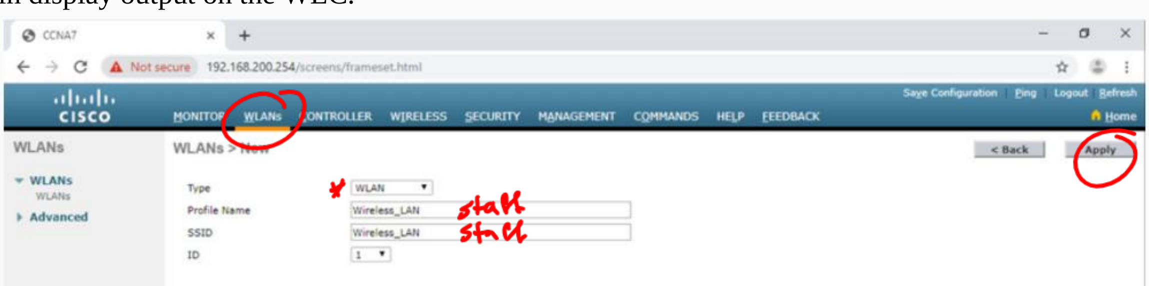
Basic WLAN configuration on the WLC includes the following steps:

1. Create the WLAN
2. Apply and Enable the WLAN
3. Select the Interface
4. Secure the WLAN
5. Verify the WLAN is Operational
6. Monitor the WLAN
7. View Wireless Client Information

*wireless controller
→ Managment
↳ Ip addres + submet mask
GUI [illegible]
→ web browser from PC,
URL:- Ip for controller.

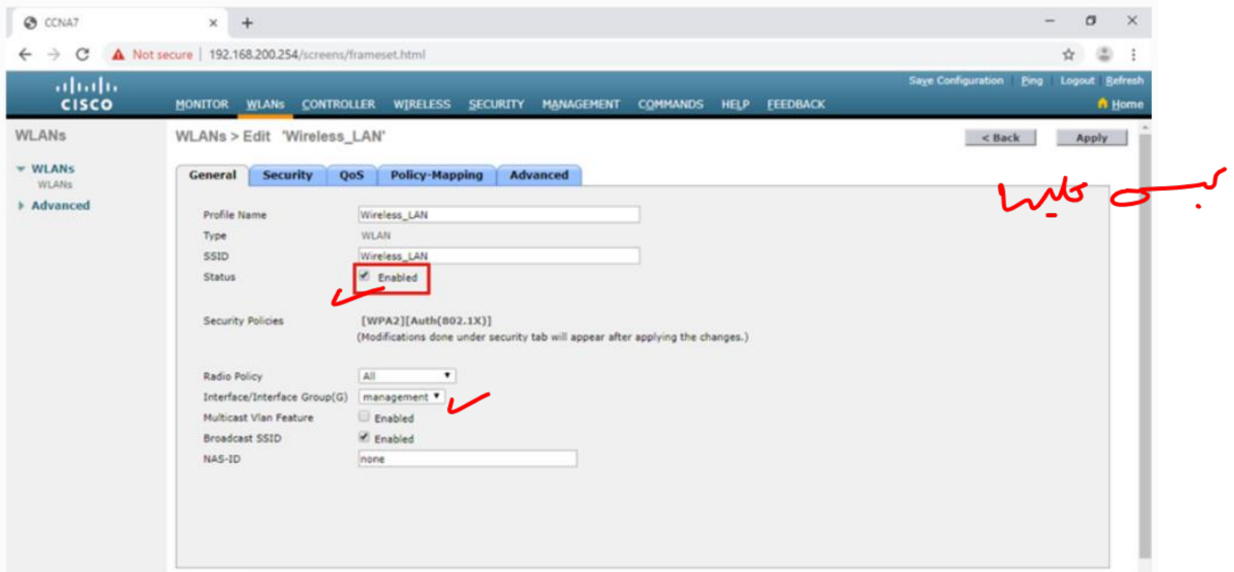
1. Create the WLAN

In the figure, the administrator is creating a new WLAN that will use **Wireless_LAN** as the name and service set identifier (SSID). The ID is an arbitrary value that is used to identify the WLAN in display output on the WLC.



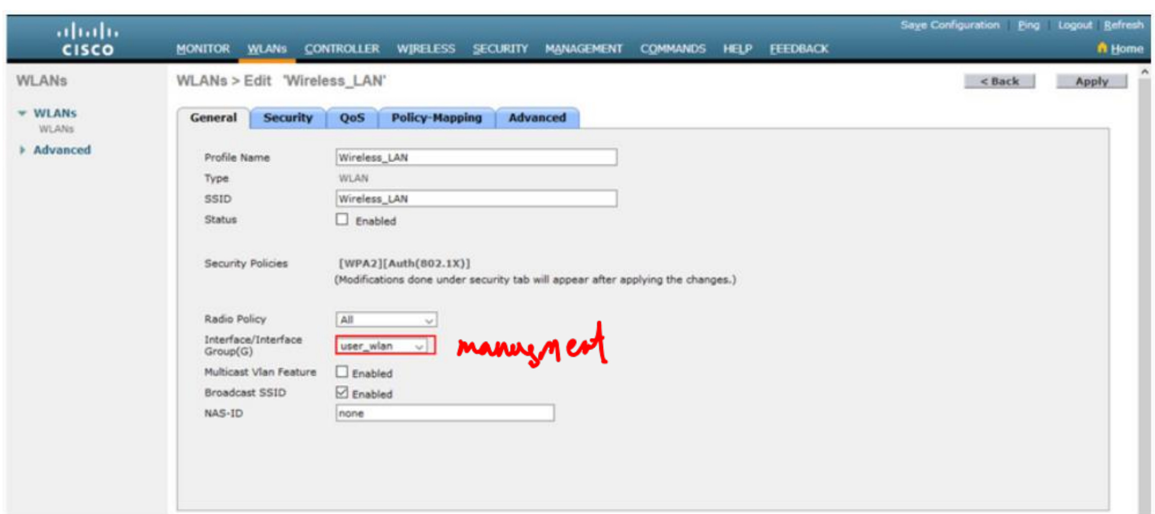
2. Apply and Enable the WLAN

After clicking **Apply**, the network administrator must enable the WLAN before it can be accessed by users, as shown in the figure. The Enable checkbox allows the network administrator to configure a variety of features for the WLAN, as well as additional WLANs, before enabling them for wireless client access. From here, the network administrator can configure a variety of settings for the WLAN including security, QoS, policies, and other advanced settings.



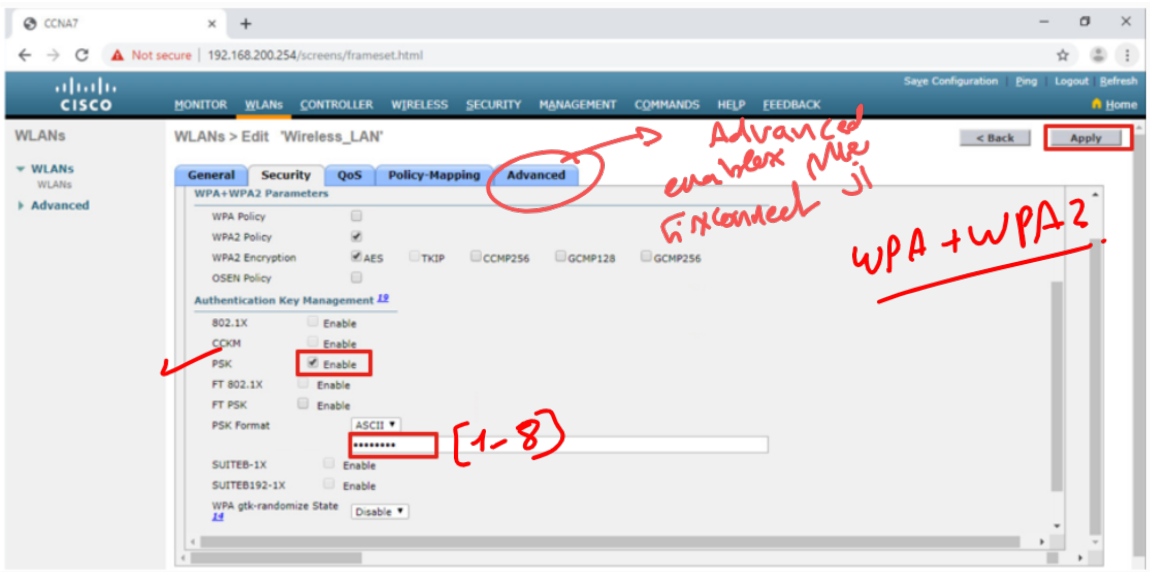
3. Select the Interface

When you create a WLAN, you must select the interface that will carry the WLAN traffic. The next figure shows the selection of an interface that has already been created on the WLC. We will learn how to create interfaces later in this module.



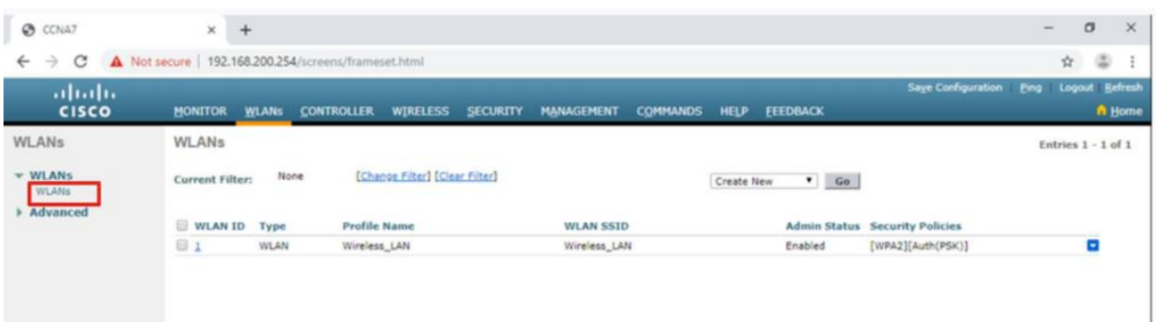
4. Secure the WLAN

Click the Security tab to access all the available options for securing the LAN. The network administrator wants to secure Layer 2 with WPA2-PSK. WPA2 and 802.1X are set by default. In the Layer 2 Security drop down box, verify that **WPA+WPA2** is selected (not shown). Click PSK and enter the pre-shared key, as shown in the figure. Then click **Apply**. This will enable the WLAN with WPA2-PSK authentication. Wireless clients that know the pre-shared key can now associate and authenticate with the AP.



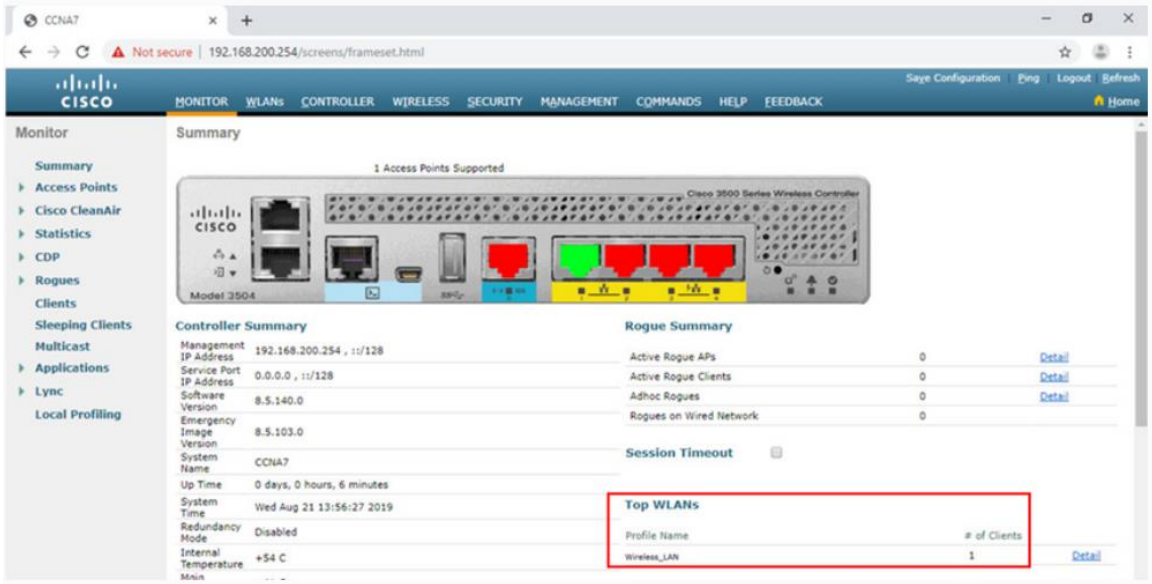
5. Verify the WLAN is Operational

Click **WLANs** in the menu on the left to view the newly configured WLAN. In the figure, you can verify that WLAN ID 1 is configured with **Wireless_LAN** as the name and SSID, it is enabled, and is using WPA2 PSK security.



6. Monitor the WLAN

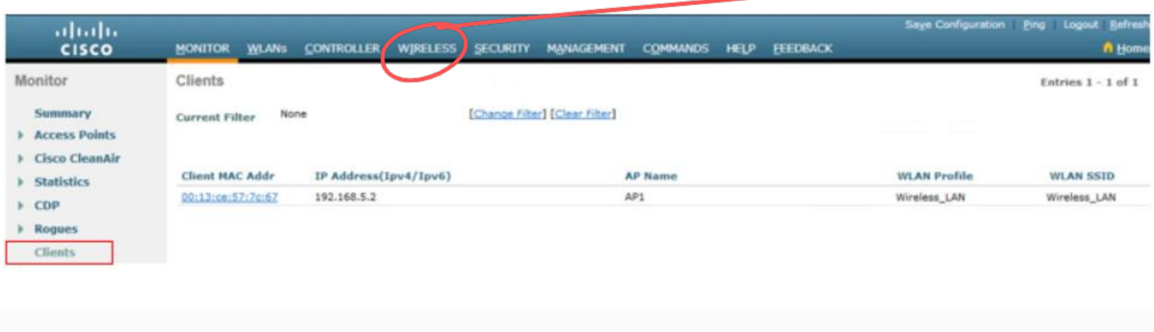
Click the **Monitor** tab at the top to access the advanced **Summary** page again. Here you can see that the **Wireless_LAN** now has one client using its services, as shown in the figure.



7. View Wireless Client Details

Click **Clients** in the left menu to view more information about the clients connected to the WLAN, as shown in the figure. One client is attached to **Wireless_LAN** through AP1 and was given the IP address 192.168.5.2. DHCP services in this topology are provided by the router.

a cen point that exists.



Procedures:

Dear students, please note that the lab problems sheet, the packet tracer activities and the practical discussion videos have been uploaded on your Microsoft Teams group. You are required to carefully study this experiment and then complete the lab sheet.

References

Cisco Networking Academy - CCNA: Switching, Routing, and Wireless Essentials.
https://www.netacad.com